# Tahreem Shahid

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## Summary

Computer Engineering student with strong experience in AI, full-stack development, and embedded systems. Skilled in building intelligent applications using LangChain, RAG pipelines, and modern web technologies. Passionate about developing scalable, real-world solutions using machine learning and agentic AI.

## Education

- **Comsats University Islamabad, Lahore.** *2022 - 2026*
  *Bachelor of Science in Computer Engineering*
  - **Coursework:** Artificial Intelligence, Machine Learning, Mobile Application Development, Data Communication and Computer Networks, Microprocessor Systems and Interfacing, Electric Circuits Analysis, Digital Logic Design.

## Experience

- **Generative AI Intern** *July 2025 – August 2025*
  *Northbay Solutions, Hybrid*
  - Utilized LangChain and Prompt Engineering for LLM integration, managing chains, memory, and templates.
  - Implemented Retrieval-Augmented Generation (RAG) to enhance AI responses with external data.
  - Used Hugging Face for NLP tasks; managed Embedding Models and Vector Databases for semantic data retrieval.
- **Frontend Development Intern** *May 2025 – July 2025*
  *Appverse Technologies, Remote*
  - Developed interactive, responsive web applications using HTML, CSS and JavaScript.
  - Enhanced UI/UX via dynamic component-based design, efficient state management, and intuitive UI features.

## Projects

- **Autonomous Indoor Disaster-Reconnaissance UGV** *Final Year Project – 2026*
  *ROS 2, MCP, AgenticROS, YOLOv8, LLMs, SLAM, FastAPI*
  - Designed an autonomous AI perception and spatial reasoning pipeline for a disaster-response UGV operating in GPS-denied environments using ROS 2, LiDAR-based SLAM, and distributed edge–cloud architecture.
  - Implemented real-time multi-stage perception with parallel YOLOv8 models for fire, smoke, and human detection (70–85% mAP), integrating LLM reasoning to generate natural-language mission reports and semantic map annotations.
  - Enabled MCP-driven agentic robot control via AgenticROS, allowing natural-language commands processed through OpenClaw and Gemini LLM, while optimizing deployment on Raspberry Pi 4 for real-time navigation without onboard GPU acceleration.
- **MCP-Integrated AI Chatbot** *March 2026 – Present*
  *LangChain, FastAPI, Vector Databases, Agentic AI*
  - Architecting an agentic document assistant using LangChain and MCP-style tools, enabling standardized communication between LLMs and external data sources with dynamic task execution.
  - Developing a FastAPI backend for tool orchestration, autonomous decision-making, and real-time context-aware query handling through modular architecture.
- **RepoRover (AI GitHub Assistant)** *February 2026*
  *LangChain, FastAPI, RAG, Vector Databases*
  - Built an AI-powered semantic search and technical Q&A system for GitHub repositories using a Retrieval-Augmented Generation (RAG) pipeline with vector database indexing.
  - Applied Agentic AI principles to transform natural language queries into repository insights and deployed an optimized backend for real-time indexing and querying.
- **Heart Disease Prediction Model** *December 2025*
  *Python, Streamlit, Scikit-learn, Pandas*
  - Developed a heart disease prediction web application using Scikit-learn with preprocessing, feature scaling, and classification modeling for improved diagnostic accuracy.
  - Designed an end-to-end data pipeline enabling real-time inference and visualization of prediction probabilities through an interactive Streamlit interface.
- **Intelligent Document Assistant** *July 2025*
  *LangChain, RAG, Vector Databases, FastAPI*
  - Developed a full-stack intelligent document assistant enabling long-document understanding through summarization, semantic comparison, and context-aware question answering.
  - Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain, prompt engineering, and vector database embeddings for efficient semantic retrieval and response generation.
  - Designed backend services for document ingestion, embedding management, and scalable querying, enabling accurate knowledge extraction from large unstructured text sources.